



Supply Chain Digitalization and Corporate Integrated Collaborative Innovation: Mechanisms, Heterogeneity, and Evidence from Chinese Listed Companies

Rong Xu¹, Lin Ren², Leilei Jiang^{3*}

¹ School of Information and Architectural Engineer, Anhui Open University, 230022 Hefei, China

² International Business School, Anhui International Studies University, 231201 Hefei, China

³ School of Economics and Management, Anhui Open University, 230022 Hefei, China

* Correspondence: Leilei Jiang (jll@ahvu.ah.cn)

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Abstract: Against the backdrop of economic globalization and rapid technological advancement, supply chain digitalization increasingly reshapes organizational collaboration and innovation patterns and has become an important driver of integrated development across supply chain networks. This study investigates whether and through what mechanisms supply chain digitalization influences corporate integrated collaborative innovation. Using panel data from Chinese A-share listed companies during 2014–2023, an empirical analysis was conducted to evaluate the effect of supply chain digitalization on integrated collaborative innovation. Fixed-effect regression models, mechanism analysis, robustness tests, and heterogeneity analysis were employed to identify both direct and indirect transmission paths. The results showed that supply chain digitalization significantly promoted corporate integrated collaborative innovation. The positive effect was transmitted through three major channels: improvements in labor productivity, increases in cost markup, and enhancement of factor allocation efficiency. The results further showed substantial heterogeneity across firms and regions. Stronger effects were observed among firms with higher levels of supply chain digitalization and those located in eastern China. The promoting effect was also more evident in producer service industries, younger firms, large-scale enterprises, and state-owned enterprises. The findings indicate that supply chain digitalization serves as an important operational mechanism for strengthening collaborative innovation capability. This study demonstrates that the coordinated development of supply chains and innovation systems contributes to more effective resource integration and sustained innovation performance. The findings provide empirical evidence for strategic decision-making related to digital transformation and offer analytical insights into the design of innovation-oriented supply chain ecosystems.

Keywords: Supply chain digitalization; Integrated collaborative innovation; Strategic analytics; Labor productivity; Cost markup; Factor allocation efficiency; Digital transformation

1 Introduction and Literature Review

Innovation has increasingly been recognized as a primary driver of high-quality economic development. As an emerging innovation paradigm, integrated collaborative innovation is characterized by resource complementarity, value co-creation, and knowledge sharing. Through the strengthening of inter-firm collaborative innovation advantages, the position of firms within global value chains is progressively enhanced. Consequently, integrated collaborative innovation has become a critical pathway through which industrial resilience can be strengthened and new quality productive forces can be cultivated. Nevertheless, the advancement of integrated collaborative innovation continues to be constrained by technological limitations, market access barriers, and institutional lock-in effects, thereby impeding both innovation upgrading at the firm level and the construction of a modernized economic system.

With the increasing diversification of consumer demand and the growing specialization of social division of labor, strategic partnerships have increasingly been sought by firms across supply chain networks. Through resource interconnection, technological collaboration, and knowledge exchange among supply chain participants, innovation chains with stronger competitive advantages are expected to be established [1]. Supply chain digitalization, which

integrates advanced technologies such as intelligent manufacturing, Internet-based platforms, and smart factories, has been driven by continuous technological innovation and has facilitated the upgrading of supply systems, thereby providing a new paradigm for integrated collaborative innovation [2]. Through enhanced organizational intelligence, greater managerial flexibility, and faster market responsiveness, supply chain digitalization has been shown to stimulate entrepreneurial innovation capabilities and strengthen the application of digital technologies among supply chain participants. As a result, deeper integration between supply chains and innovation chains has been facilitated, leading to the continuous enhancement of integrated collaborative innovation and the strengthening of firms' global competitiveness [3]. Accordingly, the acceleration of supply chain digitalization and technological innovation has become an essential approach through which integrated collaborative innovation can be promoted and entrepreneurial innovation capabilities can be reinforced [4].

Existing studies on supply chain digitalization have primarily focused on its effects on firm performance [5], labor investment efficiency [6], and green innovation [7]. In contrast, research concerning integrated collaborative innovation has largely concentrated on the relationship between digital transformation and innovation collaboration among firms. For instance, based on a sample of Chinese A-share listed companies from 2008 to 2022, Qiu and Chang [8] demonstrated that digital transformation exerts a significant positive effect on integrated collaborative innovation. In addition, the influence of digital transformation on corporate innovation has been extensively examined from multiple perspectives [9]. Using a sample of Chinese A-share listed firms during 2010–2021, Xu et al. [10] found that digital transformation facilitates green technological innovation. Duan and Zhang [11] use a sample of Chinese listed companies from 2011 to 2021 to empirically examine the impact of digital transformation on resource allocation efficiency and found that digital transformation significantly improves resource allocation efficiency. The underlying mechanisms included optimizing the structure of human capital, fostering enterprise innovation, and mitigating information asymmetry. Further evidence has been provided by Wang et al. [12], who, using panel data of Chinese manufacturing listed companies from 2012 to 2021, found that digital transformation affected the market performance and innovation performance of manufacturing enterprises, and that digital transformation effectively improved both. Through constructing an evolutionary game model of digital transformation, Liu et al. [13] demonstrated that digital transformation facilitates the formation of an innovation ecosystem, thereby ultimately affecting innovation performance. Based on organizational ambidexterity theory and using panel data from 613 Chinese listed manufacturing firms, Zhao and Gao [14] examined how the configuration types of ambidextrous innovation affect firm performance in the context of digital transformation. The study found that dual exploitation and business model leverage positively influence firm performance. Based on network embedding theory and dynamic capabilities theory, Luo et al. [15] took the panel data of Chinese A-share listed manufacturing firms from 2017 to 2022 as an example, and found that digital transformation affected the innovation resilience of manufacturing firms and played an important role in their innovation. Yu et al. [16] demonstrated that digital transformation promotes supply chain collaborative process and product innovation by strengthening inter-firm digital capabilities and collaborative mechanisms, thereby enhancing market and innovation performance. However, further expansion of both the research perspective and methodology remains warranted.

Based on the foregoing discussion, several contributions are intended to be made to the existing literature. First, a comprehensive evaluation index system for integrated collaborative innovation is systematically constructed. Using a sample of Chinese A-share listed companies from 2014 to 2023, the impact of supply chain digitalization on integrated collaborative innovation is empirically examined, thereby extending the existing literature. Second, the underlying mechanisms through which supply chain digitalization affects integrated collaborative innovation are investigated from both internal and external perspectives. By identifying the channels through which digital transformation reshapes firms' integrated collaborative innovation, a more comprehensive understanding of the relationship between supply chain digitalization and integrated collaborative innovation is provided. Third, heterogeneous effects are examined across regional, industrial, and firm-level dimensions. Differences in the impact of supply chain digitalization on integrated collaborative innovation are systematically evaluated to account for potential heterogeneity.

2 Hypothesis Development and Theoretical Analysis

2.1 Direct Effect of Supply Chain Digitalization on Integrated Collaborative Innovation

Supply chain digitalization can be conceptualized as an integrated system composed of diverse communication networks and software–hardware technologies. Through the facilitation of seamless connectivity and efficient exchange of resources both within and across organizational boundaries, a critical foundation is provided for the adoption of digital technologies and the implementation of innovation activities by firms [17]. With the support of advanced technologies such as artificial intelligence, large volumes of data, information resources, and expert knowledge can be effectively integrated, analyzed, and transformed into new knowledge assets. As a result, firms' knowledge bases can be substantially enriched, and the development of integrated collaborative innovation across supply chain networks can be effectively promoted [18–21].

According to open innovation theory, diversified supply chain systems, grounded in trust and effective

communication, facilitate the rapid exchange and dissemination of information. As a result, knowledge transfer, knowledge mobility, knowledge recombination, and knowledge application can be enhanced, generating synergistic effects that exceed the sum of individual contributions [22]. From the perspective of the mechanisms through which supply chain digitalization influences integrated collaborative innovation, digitalization can promote the development of digital innovation ecosystems, facilitate inter-organizational connectivity, and accelerate knowledge sharing across supply-side and demand-side actors, thereby creating favorable conditions for the effective implementation of integrated collaborative innovation activities [23]. On the one hand, supply chain digitalization provides essential support for the collection of critical data generated throughout procurement, production, research and development, manufacturing, and sales processes. As firms' innovation sensing capabilities are strengthened, the formation of open innovation ecosystems across organizational boundaries is facilitated [24]. Within such ecosystems, strong inter-organizational ties can be leveraged to identify structural relationships among supply chain partners, promote the establishment of innovation alliances, and deepen the level of integrated collaborative innovation.

On the other hand, entrepreneurial innovation capabilities, as a core component of organizational culture, have long been recognized as an important driver of integrated collaborative innovation. Through the application of digital technologies, supply chain digitalization expands resource-sharing networks both within and across firms and facilitates the integration of diverse innovation resources throughout supply chain networks. Consequently, the utilization advantages of heterogeneous innovation resources can be amplified, providing a favorable foundation for the cultivation of innovation-oriented entrepreneurial capabilities. Through enhanced access to, co-creation of, and sharing of innovation resources, knowledge-sharing intensity among supply chain participants can be strengthened, thereby accelerating integrated collaborative innovation. Furthermore, supply chain digitalization generates digital technology spillover effects that facilitate the establishment of stable and collaborative inter-firm relationships. Through alliance-based cooperation, entrepreneurial innovation capabilities can be further cultivated and leveraged, innovation and research and development activities can be expanded, and integrated collaborative innovation can be promoted. More importantly, supply chain digitalization broadens the scope of information nodes across supply chain networks, enabling firms to construct comprehensive innovation profiles based on diversified information sources and to acquire more extensive access to strategic knowledge resources. Through this process, entrepreneurial innovation capabilities can be continuously nurtured and stimulated, while knowledge and data resources can be transformed into innovation-oriented assets that further facilitate integrated collaborative innovation. Accordingly, the following hypothesis is proposed:

Hypothesis 1. Supply chain digitalization exerts a positive effect on corporate integrated collaborative innovation.

2.2 Indirect Effect of Supply Chain Digitalization on Integrated Collaborative Innovation

Within the broader context of the digital economy, supply chain digitalization has evolved from a strategic option into a fundamental prerequisite for firms seeking to enhance innovation capabilities, strengthen competitive advantages, and achieve long-term sustainable development. Beyond its direct contribution to integrated collaborative innovation, supply chain digitalization is expected to exert indirect effects through multiple internal and external transmission channels (Figure 1).

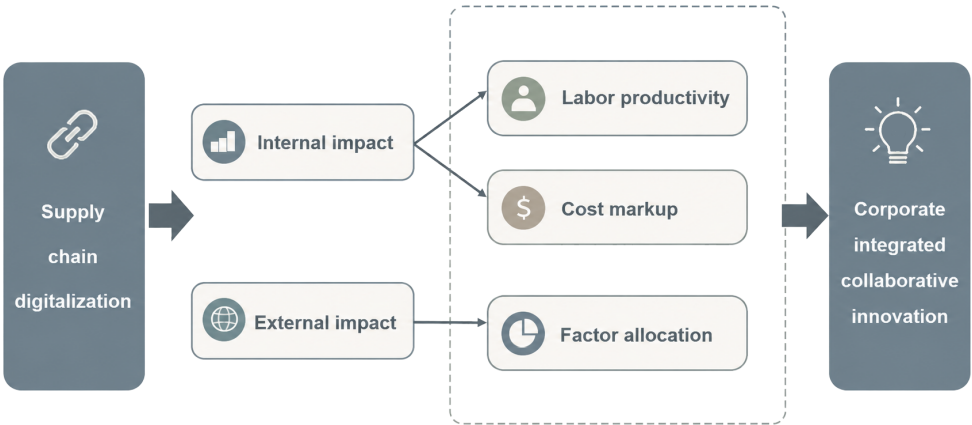


Figure 1. Impact mechanism

Within the internal mechanism, supply chain digitalization is expected to promote integrated collaborative innovation through improvements in labor productivity and increases in cost markup. With respect to labor productivity, resource-based theory suggests that sustainable competitive advantages are derived from the acquisition and effective utilization of valuable resources and capabilities. From the perspective of resource sharing, supply

chain digitalization leverages digital technologies such as big data, the Internet of Things, and blockchain to connect individuals, physical assets, and organizational units across supply chain networks. Through the digital representation and coordination of labor activities and labor relationships, information barriers among employees can be reduced, while information exchange among employees, departments, and organizational units can be strengthened. As a result, firms' capacity for value creation can be enhanced, leading to improvements in labor productivity. As labor productivity increases, firms' capabilities for acquiring information technologies are further strengthened. Consequently, the efficiency of value creation throughout production and operational processes can be enhanced, creating favorable conditions for supply chain participants to adopt advanced technologies and redefine industrial boundaries. Through these mechanisms, integrated collaborative innovation can be accelerated. With respect to cost markup, cost markup is generally defined as the ratio of product price to marginal cost. Cost markup reflects firm productivity through marginal cost and may also capture product quality through market pricing mechanisms.

Supply chain digitalization facilitates more efficient access to international market information and competitor intelligence, thereby reducing the information costs associated with upgrading within global value chains and strengthening competitive advantages. Through enhanced data transparency, organizational processes and production methods can be optimized, internal management costs can be reduced, and cost markup can be improved. More importantly, supply chain digitalization transforms traditional production and manufacturing models by enabling flexible and customized production systems that more effectively connect supply-side and demand-side participants. Through this process, value co-creation between producers and consumers can be facilitated, allowing firms to better satisfy increasingly diversified and personalized market demands. As product competitiveness is strengthened, cost markup can be further enhanced. The resulting increase in resource utilization efficiency contributes to lower marginal costs and releases additional resources that can be allocated to the development of supply chain partnerships, the strengthening of technological management capabilities, and the enhancement of innovation capacities across supply chain networks. Consequently, integrated collaborative innovation can be more effectively promoted.

From the perspective of external mechanisms, supply chain digitalization is expected to influence integrated collaborative innovation primarily through the enhancement of factor allocation. On the one hand, by leveraging machine learning techniques, supply chain digitalization enables future development prospects to be predicted on the basis of historical data. As a result, data-driven decision-making models can be established for supply chain participants, thereby mitigating the constraints imposed by bounded rationality in managerial decision-making and improving resource allocation efficiency. Through this process, resources can be more effectively concentrated across both geographical and organizational dimensions, facilitating knowledge circulation among supply chain participants. Consequently, innovation alliances can be formed more readily, and the level of integrated collaborative innovation can be enhanced. Furthermore, through the deep integration of advanced digital technologies into organizational operations, supply chain digitalization assists firms in overcoming temporal and spatial constraints and facilitates the establishment of closed-loop innovation ecosystems. Within such ecosystems, innovation resources can be allocated more efficiently, while channels for sharing commercial flows, logistics flows, and information flows among supply chain participants can be strengthened. As a result, favorable conditions are created for the effective implementation of collaborative innovation activities. On the other hand, against the backdrop of increasingly sophisticated global industrial specialization, supply chain digitalization can stimulate market expansion and interaction effects within domestic industries, thereby encouraging supply chain participants to establish more complete industrial systems and engage in higher-level innovation activities. Through these processes, innovation resources can be allocated more efficiently across both domestic and international markets. Opportunities are thereby created for firms to leverage resources from multiple markets to develop distinctive technological capabilities. Simultaneously, a dynamic innovation ecosystem characterized by demand-driven innovation and innovation-induced demand creation can be fostered throughout supply chain networks. Through the leading role of such advanced innovation ecosystems, integrated collaborative innovation can be further promoted.

Based on the foregoing theoretical analysis, the following hypothesis is proposed:

Hypothesis 2. Supply chain digitalization promotes corporate integrated collaborative innovation through improvements in labor productivity, increases in cost markup, and enhancements in factor allocation.

3 Research Design

3.1 Model Construction

Based on the foregoing theoretical analysis of the relationship between supply chain digitalization and corporate integrated collaborative innovation, and in accordance with Hypothesis 1, the following baseline model was constructed:

$$COinn_{ijt} = \alpha_0 + \alpha_1 MD_{jt} + \alpha_2 X_{ijt} + firm_i + city_i + year_i + industry_i + \varepsilon_{ijt}$$

where, $COinn_{ijt}$ denotes the level of integrated collaborative innovation of firm i in year t in place j ; MD_{jt} represents the level of supply chain digitalization in year t in place j ; X_{ijt} denotes a set of control variables; ε_{ijt}

represents the random error term; and $firm_i$, $city_i$, $year_i$, and $industry_i$ denote firm fixed effects, regional fixed effects, time fixed effects, and industry fixed effects, respectively. Particular attention was devoted to the coefficient of α_1 , which was used to measure the impact of supply chain digitalization on corporate integrated collaborative innovation.

To further test Hypothesis 2, mediation effect analysis was conducted using the causal steps approach as follows:

$$COinn_{ijt} = \alpha_0 + \alpha_1 MD_{jt} + \alpha_2 X_{ijt} + firm_i + city_i + year_i + industry_i + \varepsilon_{ijt}$$

$$M_{ijt} = \beta_0 + \beta_1 MD_{jt} + \beta_2 X_{ijt} + firm_i + city_i + year_i + industry_i + \varepsilon_{ijt}$$

$$COinn_{ijt} = \gamma_0 + \gamma_1 MD_{jt} + \gamma_2 M_{ijt} + \gamma_3 X_{ijt} + firm_i + city_i + year_i + industry_i + \varepsilon_{ijt}$$

where, M_{ijt} denotes the mediating variable. Particular attention was focused on the values of coefficients $\alpha_1, \beta_1, \gamma_1$, and γ_2 . If the coefficients were statistically significant, the existence of a partial mediation effect could be inferred.

3.2 Variable Selection

3.2.1 Dependent variable

Corporate integrated collaborative innovation ($COinn$) was used as the dependent variable. Existing research on corporate integrated collaborative innovation has largely remained at the conceptual and theoretical development stage, and a universally accepted measurement framework has yet to be established. Given the substantial variation in integrated collaborative innovation across firms and considering data availability constraints, an evaluation index system was constructed from two dimensions: integration dimension and connectivity dimension (Table 1). The level of corporate integrated collaborative innovation was then objectively evaluated using the entropy weight method.

Table 1. Evaluation index system for corporate integrated collaborative innovation

Primary Indicator	Secondary Indicator	Tertiary Indicator	Indicator Weight
Integration dimension	Resource sharing	A1: Whether a shared database, experimental facility, or communication platform has been established with collaborative partners	0.079
	Innovation output interconnectivity	A2: Natural logarithm of the number of jointly developed patents with other innovation actors plus one	0.106
	Stakeholder integration	A3: Whether equity holdings in growth enterprise market firms or venture capital companies are held	0.135
		A4: Shareholding ratio held by external institutions or investors	0.184
	Science and technology collaboration	A5: Natural logarithm of the frequency of keywords related to participation in or establishment of science and technology innovation platforms disclosed in annual reports	0.119
Connectivity dimension	Factor mobility	A6: Capital mobility—total accounts payable, prepaid accounts, advances from customers, and accounts receivable among collaborating firms, net of firm liabilities	0.079
		A7: Talent mobility—number of senior executive personnel movements among firms	0.117
		A8: Knowledge mobility—natural logarithm of patent citations and citations received for invention patents	0.092
	Commercialization of innovation outputs	A9: Ratio of new product sales revenue to patent applications within the region where the firm is located	0.089

3.2.2 Independent variable

Supply chain digitalization (MD) was used as the independent variable. Following the approach adopted in previous studies for identifying digital technologies and drawing upon related research [25], a dictionary of keywords associated with supply chain digitalization technologies was constructed. Using Python-based techniques, references to key supply chain digitalization technologies were identified from corporate annual reports. The keyword dictionary included terms related to big data, Industry 4.0, artificial intelligence, three-dimensional printing, the Internet of Things, omnichannel systems, blockchain, smart factories, unmanned aerial vehicles, digital logistics, smart supply

chains, digital supply chains, and virtual reality. Based on the identified keywords, a proxy variable for supply chain digitalization was constructed. Specifically, a value of 1 was assigned if disclosures related to supply chain digitalization technologies and associated innovation activities were identified in a firm's annual report; otherwise, a value of 0 was assigned.

3.2.3 Mediating variables

Labor productivity (*InTLP*) was used as a mediating variable. This variable was employed to measure the value created by each employee within a given period. Specifically, the indicator was calculated as the ratio of a firm's annual gross output value to its average number of employees during the year. To ensure comprehensive measurement, annual gross output value was constructed by incorporating fixed asset depreciation (total depreciation expenses), net production taxes (the difference between total taxes payable and export tax rebates receivable), employee compensation (total accrued employee remuneration), and operating surplus (operating profit plus government subsidies, gains or losses arising from fair value changes, investment income, and research and development expenditures). Following the calculation, the natural logarithm of the resulting value was taken to measure labor productivity.

Cost markup (*KP*) was used as a mediating variable. This variable was estimated using the production function approach. The firm's production function was specified as $Q_{it} = Q_{it}(X_{it}^1, \dots, X_{it}^V, K_{it}, \omega_{it})$, where K and V denote factor input and capital input, respectively, and ω_{it} represents total factor productivity. Under the assumption of cost minimization, the following Lagrangian function was constructed:

$$L(X_{it}^1, \dots, X_{it}^V, K_{it}, \lambda_{it}) = \sum_{v=1}^V P_{it}^{X^v} X_{it}^v + r_{it} K_{it} + \lambda (Q_{it} - Q_{it}(\bullet))$$

where, $P_{it}^{X^v}$ denotes the price of the variable input and r_{it} denotes the price of capital input. By applying the first-order conditions, the following expression was obtained:

$$P_{it}^{X^v} = \lambda_{it} \frac{\partial Q_{it}(\bullet)}{\partial X_{it}^v}$$

By rearranging the above expression, the following equation was obtained:

$$\frac{\partial Q_{it}(\bullet)}{\partial X_{it}^v} \frac{X_{it}^v}{Q_{it}} = \frac{P_{it}}{\lambda_{it}} \frac{P_{it}^{X^v} X_{it}^v}{P_{it} Q_{it}}$$

where, P_{it} denotes the price of the final product. According to the envelope theorem, the relationship below was derived: $\lambda_{it} = mc_{it}$. Because cost markup is defined as the ratio of product price to marginal cost, it was expressed as follows:

$$mkp_{it} = \theta_{it}^v \times (\varphi_{it}^v)^{-1}$$

where, θ_{it}^v denotes the output elasticity of the input factor X , and φ_{it}^v represents the share of the corresponding input factor in total output. Intermediate inputs are employed as the variable input factor. While φ_{it}^m was directly calculated, the estimation of θ_{it}^m required the specification and estimation of a production function. To minimize estimation bias, a Translog production function was adopted. The coefficients of the production function ($\beta_l, \beta_k, \beta_m, \beta_{ll}, \beta_{kk}, \beta_{mm}, \beta_{lk}, \beta_{lm}, \beta_{km}, \beta_{lkm}$) were estimated using the Akerberg–Caves–Frazer two-step estimation method, after which the output elasticity of intermediate inputs was derived as $\theta_{it}^m = \beta_m m_{it} + \beta_{lm} l_{it} + \beta_{km} k_{it} + \beta_{lkm} l_{it} k_{it}$.

Factor allocation (*MAF*) was used as a mediating variable. Existing studies have predominantly measured the level of factor marketization through the calculation of relative distortion coefficients in capital markets. However, such an approach tends to overlook the market-oriented allocation of other production factors, including labor, technology, and investment. To address this limitation, and drawing upon the relevant literature, a comprehensive evaluation index system was constructed to measure factor allocation efficiency from four dimensions: capital factor marketization, labor factor marketization, technology factor marketization, and investment factor marketization. The overall level of factor allocation was then evaluated using the entropy weight method. Specifically, capital factor marketization was measured by the proportion of non-state-owned investment in total fixed-asset investment. Labor factor marketization was measured by the proportion of employees working in private enterprises and self-employed businesses relative to total employment. Technology factor marketization was measured by the ratio of technology market transaction value to research and development expenditure. Investment factor marketization was measured by the proportion of investment by foreign-invested enterprises relative to gross domestic product.

3.2.4 Control variables

Following the existing literature, several firm-level characteristics were controlled for in the empirical analysis as follows:

- Firm size (*Size*). Firm size was measured as the natural logarithm of total assets plus one.
- Growth capability (*SGR*). Growth capability was measured by the annual growth rate of operating revenue.
- Operating cash flow (*CFO*). Operating cash flow was measured as the ratio of cash flow generated from operating activities to net interest-bearing debt.
- Leverage ratio (*Lev*). Leverage was measured as the ratio of total liabilities to total assets.
- Asset turnover ratio (*Tat*). Asset turnover was measured as the ratio of operating revenue to average total assets.
- Board size (*SS*). Board size was measured as the natural logarithm of the number of board directors plus one.
- Ownership type (*SOE*). Ownership type was measured using a dummy variable, where state-owned enterprises were assigned a value of 1 and non-state-owned enterprises were assigned a value of 0.

3.3 Data Sources

To examine the impact of supply chain digitalization on corporate integrated collaborative innovation, a sample of Chinese A-share listed firms on the Shanghai and Shenzhen Stock Exchanges during the period 2014–2023 was employed. Following sample selection, the data were screened according to the following criteria. First, firms operating in the financial and real estate industries were excluded. Second, firms with substantial missing information were removed. Third, firms that are financially distressed or risk-warning, along with delisted and trading-suspended ones, were excluded. Fourth, firms with an operating history of less than two years were removed from the sample. After applying these screening procedures, a final sample consisting of 2,155 firms and 18,600 firm-year observations was obtained. To mitigate the influence of outliers on the estimation results, all continuous variables were winsorized at the 1st and 99th percentiles. Firm-level data were obtained from the CSMAR Database, the Wind Database, and the China Research Data Services platform. Corporate annual reports were collected from the official websites of the Shanghai Stock Exchange and the Shenzhen Stock Exchange. Descriptive statistics for the main variables are reported in Table 2.

Table 2. Descriptive statistics

Variable	Symbol	Maximum	Minimum	Mean	Standard Deviation	Observations
Corporate integrated collaborative innovation	<i>COinn</i>	82.758	0.652	21.456	9.152	18600
Supply chain digitalization	<i>MD</i>	1.000	0.000	0.401	0.122	18600
Labor productivity	<i>InTLP</i>	15.632	10.426	12.696	0.695	18600
Cost markup	<i>KP</i>	1.245	0.522	0.748	0.109	18600
Factor allocation	<i>MAF</i>	1.501	0.385	0.796	0.355	18600
Firm size	<i>Size</i>	26.965	18.542	21.362	1.329	18600
Growth capability	<i>SGR</i>	0.402	-0.542	0.090	0.076	18600
Operating cash flow	<i>CFO</i>	76.529	-3.628	5.478	10.233	18600
Leverage ratio	<i>Lev</i>	0.967	0.055	0.405	0.202	18600
Asset turnover ratio	<i>Tat</i>	2.732	0.115	0.718	0.449	18600
Board size	<i>SS</i>	2.236	1.395	1.210	0.185	18600
Ownership type	<i>SOE</i>	1.000	0.000	0.456	0.328	18600

4 Empirical Results and Analysis

4.1 Baseline Regression

Regression analysis based on Model (1) was conducted to test Hypothesis 1. The estimation results are reported in Table 3. As shown in the table, when no control variables are included, the coefficient of *MD* is estimated to be 0.852 and is statistically significant at the 1% level. After the inclusion of control variables and the sequential incorporation of city, industry, firm, and time fixed effects, the coefficient of *MD* remains positive and statistically significant. These findings suggest that improvements in the level of supply chain digitalization significantly promote corporate integrated collaborative innovation, thereby providing support for Hypothesis 1. A plausible explanation is that supply chain digitalization facilitates the reallocation of resources across supply chain networks through the application of digital technologies. By reducing information asymmetry among firms, supply chain digitalization enhances both the incentives and the efficiency of integrated collaborative innovation activities.

Table 3. Baseline regression results

Variables	(1) <i>COinn</i>	(2) <i>COinn</i>	(3) <i>COinn</i>	(4) <i>COinn</i>
<i>MD</i>	0.852*** (5.46)	0.635*** (4.85)	0.557*** (4.30)	0.321*** (3.59)
<i>Size</i>		0.211*** (12.35)	0.242*** (12.20)	0.258*** (12.86)
<i>SGR</i>		0.322*** (4.56)	0.315*** (4.45)	0.326*** (4.59)
<i>CFO</i>		2.102*** (16.52)	1.859*** (16.42)	1.837*** (17.59)
<i>Lev</i>		-0.756*** (-13.27)	-0.695*** (-11.56)	-0.695*** (-12.32)
<i>Tat</i>		0.157*** (2.98)	0.169*** (3.45)	0.458*** (2.85)
<i>SS</i>		0.257*** (19.45)	0.248*** (21.53)	0.211*** (19.27)
<i>SOE</i>		0.162*** (9.62)	0.157*** (9.47)	0.155*** (11.32)
Constant	8.635*** (3.68)	2.625*** (8.37)	2.478*** (9.05)	1.901*** (4.57)
City fixed effects	No	No	No	Yes
Industry fixed effects	No	No	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.3682	0.4852	0.5827	0.6372
Observations	18600	18600	18600	18600

Note: Cluster-robust t -statistics at the firm level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The same notation applies to all subsequent tables. *COinn* = corporate integrated collaborative innovation; *MD* = supply chain digitalization; *Size* = firm size; *SGR* = growth capability; *CFO* = operating cash flow; *Lev* = leverage ratio; *Tat* = asset turnover ratio; *SS* = board size; *SOE* = ownership type.

4.2 Robustness and Endogeneity Tests

4.2.1 Robustness tests

To ensure the robustness of the findings, several robustness tests were conducted. First, the measurement of the dependent variable was replaced. Specifically, corporate integrated collaborative innovation was re-estimated using the entropy-weighted Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method, and the regression analysis was re-performed. Second, to mitigate potential concerns regarding reverse causality, both *MD* and *COinn* were lagged by one period and subsequently incorporated into the regression analysis. As reported in columns (1) and (2) of the table, only minor changes are observed in the estimated coefficients of the core explanatory variable. Moreover, the statistical significance of the coefficients remains largely unchanged. These results indicate that the positive relationship between supply chain digitalization and integrated collaborative innovation remains robust.

4.2.2 Endogeneity analysis

Potential endogeneity concerns may arise due to the reciprocal relationship between supply chain digitalization and corporate integrated collaborative innovation. In addition, omitted variables associated with firms' innovation activities may also introduce endogeneity bias into the estimation results. To mitigate these concerns, an instrumental variable approach was employed, with the initial level of *MD* set as the instrumental variable. The rationale is that the initial level of digitalization may influence the subsequent trajectory of digital transformation. At the same time, it is expected to exert no direct effect on current integrated collaborative innovation other than through its impact on supply chain digitalization, thereby satisfying the exogeneity and related conditions. Based on this specification, instrumental variable estimation was conducted. The results reported in column (3) of Table 4 indicate that supply chain digitalization significantly promotes corporate integrated collaborative innovation.

Table 4. Robustness and endogeneity test results

Variables	Alternative Measure (1) <i>COinn</i>	One-Period Lagged Variables (2) <i>COinn</i>	Instrumental Variables (3) <i>COinn</i>
<i>MD</i>	0.698*** (4.54)	0.621*** (4.48)	0.584*** (5.01)
<i>Size</i>	0.228*** (11.54)	0.274*** (12.84)	0.312*** (12.63)
<i>SGR</i>	0.336*** (4.58)	0.362*** (4.52)	0.380*** (4.58)
<i>CFO</i>	2.096*** (13.62)	2.084*** (14.52)	2.072*** (14.15)
<i>Lev</i>	-0.693*** (-12.52)	-0.628*** (-11.52)	-0.605*** (-12.81)
<i>Tat</i>	0.173*** (2.99)	0.60*** (3.52)	0.168*** (3.84)
<i>SS</i>	0.275*** (22.33)	0.359*** (20.05)	0.384*** (20.47)
<i>SOE</i>	0.58*** (11.25)	0.172*** (10.85)	0.181*** (11.54)
Constant	2.754*** (9.85)	2.84*** (10.60)	2.025*** (9.52)
City fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes
Adjusted R^2	0.3625	0.5148	0.5348
Observations	18600	16740	18600

Note: *COinn* = corporate integrated collaborative innovation; *MD* = supply chain digitalization; *Size* = firm size; *SGR* = growth capability; *CFO* = operating cash flow; *Lev* = leverage ratio; *Tat* = asset turnover ratio; *SS* = board size; *SOE* = ownership type.

4.3 Mechanism Analysis

The mediating effect of labor productivity was first examined. As reported in column (1) of Table 5, the coefficient of supply chain digitalization is estimated at 0.032 and is statistically significant at the 5% level, indicating that a one-unit increase in supply chain digitalization is associated with a 0.032-unit increase in labor productivity. The results presented in column (2) show that the coefficient of *InTLP* is 3.785 and is significant at the 1% level. At the same time, the coefficient of *MD* remains positive and significant at 0.192. These findings suggest that labor productivity serves as a partial mediating variable in the relationship between supply chain digitalization and integrated collaborative innovation. A plausible explanation is that the adoption and diffusion of digital technologies across supply chain networks alter firms' human resource allocation structures and improve workforce productivity, thereby facilitating integrated collaborative innovation.

The mediating effect of cost markup was subsequently examined. According to the results reported in column (3) of Table 5, the coefficient of *MD* is estimated at 0.145 and is statistically significant at the 5% level, indicating that a one-unit increase in supply chain digitalization is associated with a 0.145-unit increase in cost markup. As shown in column (4), the coefficient of *KP* is estimated at 0.627 and is significant at the 1% level, while the coefficient of *MD* remains significantly positive at 0.254. These findings indicate that cost markup functions as an effective partial mediating variable, suggesting that supply chain digitalization promotes integrated collaborative innovation through improvements in firms' cost markup. This effect may be attributed to the restructuring of organizational processes induced by digital transformation. As organizational operations become increasingly intelligent and technology-driven, thereby contributing to higher cost markups and creating favorable conditions for integrated collaborative innovation.

Finally, the mediating effect of factor allocation was examined. The results reported in column (5) of Table 5 indicate that the coefficient of *MD* is 0.029 and is statistically significant at the 10% level, suggesting that a one-unit increase in supply chain digitalization is associated with a 0.029-unit increase in factor allocation efficiency. As shown in column (6), the coefficient of *MAF* efficiency is estimated at 1.801 and is significant at the 1% level, while the coefficient of *MD* remains significantly positive at 0.325. These results demonstrate that factor allocation serves as an effective external mediating mechanism linking supply chain digitalization and integrated collaborative

innovation. A potential explanation is that advanced digital technologies facilitate the circulation and coordination of production factors across firms and throughout production and distribution processes. The enhancement of factor allocation provides a solid foundation for integrated collaborative innovation.

Table 5. Mechanism analysis results

Variables	(1) <i>InTLP</i>	(2) <i>COinn</i>	(3) <i>KP</i>	(4) <i>COinn</i>	(5) <i>MAF</i>	(6) <i>COinn</i>
<i>MD</i>	0.032** (3.52)	0.192** (2.10)	0.145** (2.28)	0.254*** (2.97)	0.029* (1.89)	0.352*** (3.84)
<i>InTLP</i>		3.785*** (20.52)				
<i>KP</i>				0.627*** (10.95)		
<i>MAF</i>						1.801*** (8.80)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Constant	0.241*** (7.84)	0.585*** (6.12)	0.501*** (7.51)	0.408*** (8.93)	0.411*** (9.47)	0.605*** (10.52)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.3521	0.3284	0.3840	0.3621	0.3528	0.3009
Observations	18600	18600	18600	18600	18600	18600

Note: *COinn* = corporate integrated collaborative innovation; *MD* = supply chain digitalization; *InTLP* = labor productivity; *KP* = cost markup; *MAF* = factor allocation.

4.4 Heterogeneity Analysis

4.4.1 Heterogeneity by the level of supply chain digitalization

Building upon the preceding analysis, the sample was divided into two groups according to the median value of the supply chain digitalization indicator: a high-supply chain digitalization group and a low-supply chain digitalization group. The corresponding regression results are reported in columns (1) and (2) of Table 6. The results indicate that the coefficient of *MD* in the high-supply chain digitalization group is estimated at 0.405 and is statistically significant at the 10% level. In contrast, the estimated coefficient in the low-supply chain digitalization group is 0.280 and fails to reach conventional levels of statistical significance. These findings suggest that the positive relationship between supply chain digitalization and integrated collaborative innovation is more pronounced among firms characterized by a higher level of supply chain digitalization. A plausible explanation is that firms with more advanced levels of supply chain digitalization are able to achieve deeper integration of digital technologies throughout supply chain networks. As digital infrastructures and inter-organizational connectivity are strengthened, the effectiveness of integrated collaborative innovation is further enhanced.

Table 6. Heterogeneity analysis by supply chain digitalization level and region

Variables	(1) High-Supply Chain Digitalization Group	(2) Low-Supply Chain Digitalization Group	(3) Eastern Region	(4) Central and Western Regions
<i>MD</i>	0.405* (1.75)	0.280 (1.02)	0.596*** (3.87)	0.205 (1.08)
Control variables	Yes	Yes	Yes	Yes
Constant	1.822*** (12.98)	1.931*** (10.19)	1.826*** (11.48)	1.779*** (10.34)
City fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.6358	0.5962	0.6852	0.6857
Observations	10080	8520	9640	8960

Note: *MD* = supply chain digitalization.

4.4.2 Regional heterogeneity

From a geographical perspective, the heterogeneous effects of supply chain digitalization on integrated collaborative innovation were further examined. Based on the provincial location of firms' operating cities, the sample was divided into two groups: the eastern region and the central and western regions. The corresponding estimation results are reported in columns (3) and (4) of Table 6. As shown in Table 6, the estimated coefficient of *MD* for firms located in the central and western regions is 0.205, although the coefficient fails to reach conventional levels of statistical significance. In contrast, the coefficient for firms located in the eastern region is estimated at 0.596 and is statistically significant at the 1% level. These findings indicate that the positive relationship between supply chain digitalization and integrated collaborative innovation is considerably stronger among firms located in the eastern region. A plausible explanation is that the eastern region generally possesses superior labor resources, capital availability, digital infrastructure, and innovation-supporting institutions. Such advantages provide more favorable conditions for the implementation of supply chain digitalization and the facilitation of integrated collaborative innovation.

4.4.3 Industry heterogeneity

From the perspective of industry characteristics, the heterogeneous effects of supply chain digitalization on integrated collaborative innovation were further examined across different sectors. Based on firms' industry classifications, the sample was divided into the manufacturing sector and the service sector, and separate regressions were conducted. The corresponding results are reported in columns (1) and (2) of Table 7. The results indicate that the estimated coefficient of *MD* for manufacturing firms is 0.128 and does not reach conventional levels of statistical significance. In contrast, the coefficient for service-sector firms is estimated at 0.658 and is statistically significant at the 1% level. These findings suggest that the positive relationship between supply chain digitalization and integrated collaborative innovation is considerably stronger among service-sector firms than among manufacturing firms.

To further investigate heterogeneity within the service sector, service firms were divided into producer services and consumer services, and separate regressions were estimated. The results reported in columns (3) and (4) of Table 7 show that the coefficients of supply chain digitalization are 0.805 and 0.612, respectively, and are statistically significant at the 1% and 5% levels. These findings indicate that the effect of supply chain digitalization on integrated collaborative innovation is more pronounced among producer-service firms than among consumer-service firms. A plausible explanation is that service industries are primarily engaged in the creation and sales of intangible assets, making them particularly compatible with digital technologies. As a result, digital transformation facilitates the development of new products, platforms, business models, and service ecosystems, thereby creating favorable conditions for integrated collaborative innovation. This effect is especially evident in producer services, where the marginal benefits of digital technologies can be more effectively leveraged through digital transformation. By promoting business model innovation and upgrading, supply chain digitalization contributes more substantially to integrated collaborative innovation.

Table 7. Industry heterogeneity analysis

Variables	(1) Manufacturing Sector	(2) Service Sector	(3) Producer Services	(4) Non-Producer Services
<i>MD</i>	0.128 (0.98)	0.658*** (3.87)	0.805*** (3.85)	0.612** (2.40)
Control variables	Yes	Yes	Yes	Yes
Constant	0.215*** (12.05)	0.188*** (4.85)	0.159*** (3.94)	0.206*** (3.66)
City fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.5243	0.5897	0.5578	0.5687
Observations	13280	5320	2506	2814

Note: *MD* = supply chain digitalization.

4.4.4 Firm age heterogeneity

The sample was divided into young firms and mature firms according to the median value of firm age. The corresponding regression results are reported in columns (1) and (2) of Table 8. The results indicate that the estimated coefficient of *MD* for mature firms is 0.140 and does not reach conventional levels of statistical significance. In contrast, the coefficient for young firms is estimated at 0.251 and is statistically significant at the 5% level. These

findings suggest that the positive relationship between supply chain digitalization and integrated collaborative innovation is more pronounced among young firms than among mature firms. A plausible explanation is that younger firms are generally closer to the digital transformation stage and tend to be more responsive to emerging digital technologies and external technological changes, thereby enhancing integrated collaborative innovation.

Table 8. Heterogeneity analysis by firm age, firm size, and ownership type

Variables	(1) Mature Firms	(2) Young Firms	(3) Large Firms	(4) Small Firms	(5) Non-State-Owned Enterprises	(6) State-Owned Enterprises
<i>MD</i>	0.140 (0.87)	0.251* (1.89)	0.425*** (3.18)	0.150 (0.72)	0.224 (1.01)	0.362*** (3.87)
Constant	0.206*** (7.56)	0.185*** (7.43)	0.309*** (15.24)	0.189*** (6.42)	0.273*** (8.46)	0.213*** (10.28)
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.5421	0.5682	0.5527	0.6128	0.6203	0.6785
Observations	9500	9100	9260	9340	8360	10240

Note: *MD* = supply chain digitalization.

4.4.5 Firm size heterogeneity

To further investigate whether the impact of supply chain digitalization differs across firms of varying sizes, the sample was divided into large firms and small firms according to the median value of firm size. The corresponding estimation results are reported in columns (3) and (4) of Table 8. The results show that the coefficient of *MD* for large firms is estimated at 0.425 and is statistically significant at the 1% level. By contrast, the coefficient for small firms is estimated at 0.150 and fails to reach conventional levels of statistical significance. These findings indicate that the positive effect of supply chain digitalization on integrated collaborative innovation is substantially stronger among large firms. A potential explanation is that large firms generally possess superior financial resources, human capital, and technological capabilities. These advantages enable firms to capitalize more effectively on digital transformation opportunities. In contrast, small firms are often constrained by limited financial resources, labor availability, and technological capabilities. As a result, innovation activities are more likely to rely on basic information technologies, resulting in relatively lower sensitivity to supply chain digitalization.

4.4.6 Ownership heterogeneity

From the perspective of ownership structure, the heterogeneous effects of supply chain digitalization on integrated collaborative innovation were further examined. The sample was divided into state-owned enterprises and non-state-owned enterprises, and separate regressions were estimated. The corresponding results are reported in columns (5) and (6) of Table 8. The results indicate that the estimated coefficient of *MD* for non-state-owned enterprises is 0.224 and does not reach conventional levels of statistical significance. In contrast, the coefficient for state-owned enterprises is estimated at 0.362 and is statistically significant at the 1% level. These findings suggest that the positive effect of supply chain digitalization on integrated collaborative innovation is substantially stronger among state-owned enterprises than among non-state-owned enterprises. A plausible explanation is that state-owned enterprises often play a prominent role in digital infrastructure construction, which facilitates the adoption and application of advanced digital technologies, thereby enabling the benefits of supply chain digitalization to be more effectively translated into integrated collaborative innovation. By contrast, non-state-owned enterprises are more likely to face financing constraints and market-entry barriers. As a result, fewer opportunities may be available for undertaking technological innovation via extensive policy advantages and resource factors, which may weaken the effectiveness of supply chain digitalization in promoting integrated collaborative innovation.

5 Conclusions and Policy Implications

Against the backdrop of the deepening development of the digital economy, supply chain digitalization has become increasingly critical for integrated collaborative innovation among firms. Using a sample of Chinese A-share listed firms on the Shanghai and Shenzhen Stock Exchanges from 2014 to 2023, this study examines the impact of supply chain digitalization on integrated collaborative innovation. The results indicate that supply chain digitalization consistently promotes integrated collaborative innovation. The mechanism analysis further reveals that integrated collaborative innovation is facilitated through improvements in labor productivity, cost markup, and factor allocation. The heterogeneity analysis demonstrates that the positive effect of supply chain digitalization is more pronounced among firms characterized by higher levels of supply chain digitalization, firms located in the

eastern region, service-sector firms—particularly those operating in producer services—young firms, large firms, and state-owned enterprises.

Given the importance of supply chain digitalization and integrated collaborative innovation for innovation-driven development in the real economy, several policy implications can be derived below.

First, entrepreneurial spirit should be fostered to promote the deep integration of the supply chain and the innovation chain. The cultivation of entrepreneurial spirit is conducive to stimulating firms' innovation potential, strengthening innovation-oriented strategic capabilities, and encouraging the continuous exploration of emerging market opportunities, thereby accelerating the process of integrated collaborative innovation. Accordingly, collaborative development and resource-sharing principles should be embraced, while entrepreneurial spirit should be leveraged to facilitate the deep integration of supply chains and innovation chains. On the one hand, greater emphasis should be placed on expanding diversified supply chain information channels and establishing information-sharing platforms and innovation ecosystems among supply chain participants. Such efforts would provide a favorable foundation for the development of entrepreneurial spirit. In addition, the advantages associated with digital technologies should be fully utilized to broaden the scope of digital collaboration among supply chain partners, strengthen the integration of internal and external resources, and promote the joint cultivation of entrepreneurial spirit across multiple stakeholders, thereby deepening the integration between supply chains and innovation chains. On the other hand, active responses should be provided to the digital transformation needs and strategic concerns of supply chain partners. Through digital platforms, opportunities for cultivating entrepreneurial spirit characterized by innovation and exploration should be continuously identified and expanded. Long-term cooperative relationships should also be established among supply chain participants to strengthen entrepreneurial spirit, accelerate the integration of supply chains and innovation chains, and gradually enhance firms' integrated collaborative innovation capabilities.

Second, the scope of factor utilization should be expanded to enhance internal and external resource sharing. Under the transformative influence of supply chain digitalization, firms' innovation paradigms have undergone profound changes. The enhancement of integrated collaborative innovation through the effective utilization of production factors, including labor, capital, and technology, has become increasingly important for achieving sustainable growth. Accordingly, greater efforts should be devoted to expanding the scope of factor utilization and strengthening resource sharing both within and across organizations. Potential innovation partners should be actively identified, and collaborative innovation activities should be promoted through the coordinated utilization of key innovation factors, including labor, capital, and technology. A unified digital platform should be established by leveraging the factor allocation environment created by supply chain digitalization. Such a platform would provide an open and collaborative space for the sharing of innovation resources among firms and facilitate the effective implementation of integrated collaborative innovation activities. In addition, the intrinsic characteristics of supply chain digitalization should be fully leveraged. By utilizing the permeable and integrative nature of digital technologies, technological innovation scenarios across supply chain participants can be interconnected, thereby extending the scope of factor utilization, expanding internal and external resource reserves, enriching firms' resource pools, strengthening resource-sharing capabilities, and enhancing the vitality of integrated collaborative innovation.

Third, innovation ecosystems should be established based on firms' resource endowment advantages. The empirical findings indicate that the effects of supply chain digitalization on integrated collaborative innovation vary across regions, industries, and firm characteristics. Accordingly, innovation ecosystems should be developed in accordance with firms' resource endowment advantages to fully leverage the benefits of integrated collaborative innovation. On the one hand, firms located in the eastern region, producer-service industries, young firms, large firms, and state-owned enterprises should be encouraged to assume leadership and demonstration roles. Innovation alliances characterized by digital technology sharing, knowledge exchange, risk sharing, mutual benefit, and collaborative development should be established. Through industrial collaborative innovation, differentiated innovation ecosystems can be cultivated to facilitate innovation-driven development among firms located in the central and western regions, manufacturing firms, mature firms, small firms, and non-state-owned enterprises. On the other hand, firms possessing stronger innovation capabilities should further accelerate digital transformation efforts. Through the establishment of digital transformation support mechanisms, less-advantaged firms can be assisted in enhancing their integrated collaborative innovation capabilities. Furthermore, intellectual property protection mechanisms should be effectively utilized to promote both technology acquisition and technology diffusion. The participation of less-advantaged firms in broader innovation ecosystems should also be encouraged, thereby maximizing the positive effects of supply chain digitalization and further enhancing the level of integrated collaborative innovation.

Author Contributions

Conceptualization, R.X. and L.R.; methodology, L.L.J.; software, R.X.; validation, L.R. and L.L.J.; formal analysis, L.L.J.; investigation, R.X.; resources, L.R.; data curation, L.R.; writing—original draft preparation, R.X.; writing—review and editing, L.L.J.; visualization, L.R.; supervision, L.L.J.; project administration, L.L.J.; funding acquisition, R.X. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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